



Rate Plans Module v1

Runbook, Technical Documentation & Test Cases



Module Overview

Purpose

The **Rate Plans module** defines and manages **room pricing structures** for a hotel (tenant). It serves as the **pricing foundation** for:

- Reservations v2 (price attachment)
 - Billing / Folio calculations
 - Future seasonal and dynamic pricing
-

Scope (v1)

- Flat base pricing per rate plan
 - Validity period (`validFrom` → `validTo`)
 - Strict tenant isolation
 - ADMIN-controlled mutations
 - Read access for STAFF
-



Core Concepts

Tenant Model

- **Tenant = Hotel**
 - Each Rate Plan belongs to **exactly one tenant**
 - Cross-tenant access is **strictly forbidden**
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Rate Plan Lifecycle

- No lifecycle transitions in v1
- Status is static:
- `ACTIVE`
- `INACTIVE`

Future versions may add: - Auto-expiry - Seasonal overrides - Yield / demand-based pricing

Entity Model

RatePlan Entity

Field	Type	Notes
id	UUID	Primary key
code	string	Unique per tenant
name	string	Display name
description	string null	Optional
basePrice	decimal(10,2)	Base nightly rate
validFrom	date	Start of validity
validTo	date	End of validity
status	enum	ACTIVE / INACTIVE
tenant	FK → Tenant	Mandatory
createdAt	timestamp	Auto
updatedAt	timestamp	Auto

Database Constraints

- Unique index: (tenant_id, code)
- Foreign key: rate_plans.tenant_id → tenants.id
- Delete rule: RESTRICT

Business Rules

Creation Rules

- validFrom < validTo (strict)
- Tenant must exist
- Code must be unique per tenant
- Only ADMIN can create

Update Rules

- Only ADMIN can update
- Partial updates allowed
- Validity rules re-validated
- Status can be toggled

Access Rules (RBAC)

Action	ADMIN	STAFF
Create rate plan	✓	✗
List rate plans	✓	✓
Get rate plan by ID	✓	✓
Update rate plan	✓	✗

API Endpoints

Base Path

/rate-plans

Create Rate Plan

POST /rate-plans?tenantCode=TENANT_CODE

Auth: JWT (ADMIN)

Request Body

```
{
  "code": "BAR",
  "name": "Best Available Rate",
  "description": "Standard public rate",
  "basePrice": 250.00,
  "validFrom": "2026-02-01",
  "validTo": "2026-12-31"
}
```

List Rate Plans

GET /rate-plans?tenantCode=TENANT_CODE

Auth: JWT (ADMIN, STAFF)

Get Rate Plan by ID

GET `/rate-plans/:id?tenantCode=TENANT_CODE`

Auth: JWT (ADMIN, STAFF)

Update Rate Plan

PATCH `/rate-plans/:id?tenantCode=TENANT_CODE`

Auth: JWT (ADMIN)

Request Body

```
{
  "name": "Best Available Rate - Updated",
  "basePrice": 275.00
}
```

Error Handling

Scenario	HTTP
Missing / invalid JWT	401
Forbidden role	403
Missing tenantCode	400
Tenant not found	404
Rate plan not found	404
Invalid date range	400

Test Cases (curl + jq)

Test Case 1 – Create (ADMIN)

```
curl -s -X POST http://localhost:3000/rate-plans?tenantCode=SUNU_HOTEL
-H "Authorization: Bearer $ADMIN_TOKEN"
-H "Content-Type: application/json"
-d '{
  "code": "BAR",
  "name": "Best Available Rate",
  "basePrice": 250,
```

```
"validFrom": "2026-02-01",  
"validTo": "2026-12-31"  
' | jq
```

Test Case 2 – List

```
curl -s http://localhost:3000/rate-plans?tenantCode=SUNU_HOTEL  
-H "Authorization: Bearer $ADMIN_TOKEN" | jq
```

Test Case 3 – Get by ID

```
curl -s http://localhost:3000/rate-plans/{RATE_PLAN_ID}?  
tenantCode=SUNU_HOTEL  
-H "Authorization: Bearer $ADMIN_TOKEN" | jq
```

Test Case 4 – Update (ADMIN)

```
curl -s -X PATCH http://localhost:3000/rate-plans/{RATE_PLAN_ID}?  
tenantCode=SUNU_HOTEL  
-H "Authorization: Bearer $ADMIN_TOKEN"  
-H "Content-Type: application/json"  
-d '{ "basePrice": 275 }' | jq
```

Test Case 5 – Update (STAFF → Forbidden)

```
curl -s -X PATCH http://localhost:3000/rate-plans/{RATE_PLAN_ID}?  
tenantCode=SUNU_HOTEL  
-H "Authorization: Bearer $STAFF_TOKEN"  
-H "Content-Type: application/json"  
-d '{ "basePrice": 999 }' | jq
```

Expected:

```
{ "statusCode": 403 }
```

Module Status

Rate Plans v1 is LOCKED

- Code implemented
- Runtime tested
- RBAC verified
- Documentation complete